Domain Resources

Domain Lexicon

Entity Taxonomy

Co-occurrence Rules

Synthetic Bootstrap Data

LLM Context Synthesis

Constrain LLM to generate academic sentences

Rule-based Validation

Deterministic String Matching & Boundary Check

Synthetic Dataset D_syn

BERT Boundary Model

Input tokens: [CLS] | arsenian | pyrite | occurs | in | quartz | veins | [SEP]

BERT encoder: Transformer layer 1 ... Transformer layer L

Contextual vectors: $h_{[CLS]}$ h_1 h_2 h_3 h_4 h_5 h_6 $h_{[SEP]}$

	[CLS]	arsenian	pyrite	occurs	in	quartz	veins	[SEP]
Start head	0.03	0.92	0.34	0.01	0.02	0.90	0.25	0.01
End head	0.03	0.06	0.84	0.02	0.02	0.45	0.75	0.01

Type head removed

Candidate Span: arsenian pyrite; quartz veins

Self Training

Sentence context

Candidate Span

... arsenian pyrite ...

Start End

Unlabeled Chunks ...

LLM Hierarchical Typing

Coarse Type	Medium Type	Fine Type
Composition	Mineral	SulfideMineral

Typed Entity Mentions